AP CALCULUS AB

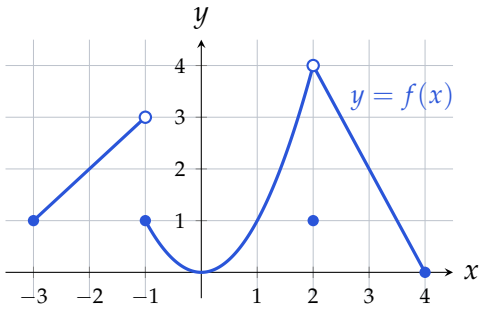
Unit 1 • Self Check 1.1 • Lessons 1-1 and 1-2 • Answer key

Self Check 1.1 • Lessons 1-1 and 1-2

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



Question 1

NO CALCULATOR

Using the graph of f, $\lim_{x \to -1^+} f(x) =$

(A) 3
(B) 1
(C) 0
(D) It does not exist.

Solution

Answer: (B).

Question 2

NO CALCULATOR

Using the graph of f, $\lim_{x \to 3} f(x) =$

(A) 1
(B) 3
(C) 2
(D) 4

Solution

Answer: (C). $8 - 6 = 2$.

Question 3

NO CALCULATOR

$$\lim_{x\to -2} \frac{x^2-4}{x+2} =$$

(A) 0
(B) -4
(C) 4
(D) It does not exist.

Solution
Answer: (B).

Question 4

NO CALCULATOR

$$\lim_{x\to 1} \frac{x^2+2x-3}{x-1} =$$

(A) 4
(B) 0
(C) 2
(D) It does not exist.

Solution
Answer: (A). $\frac{(x-1)(x+3)}{x-1} = x+3 \to 4.$

Question 5 (Free response)

NO CALCULATOR

Find $\lim_{x\to 0} \frac{x^2+5x}{x}$, $\lim_{x\to -3} \frac{x^2-9}{x+3}$, and $\lim_{x\to 2} j(x)$ for $j(x) = \begin{cases} x^2, & x < 2 \\ 6-x, & x \geq 2. \end{cases}$

Solution
$x+5 \to 5$; $x-3 \to -6$; left 4, right 4: the limit is 4.